

Lecture 12

Optimization for Inference

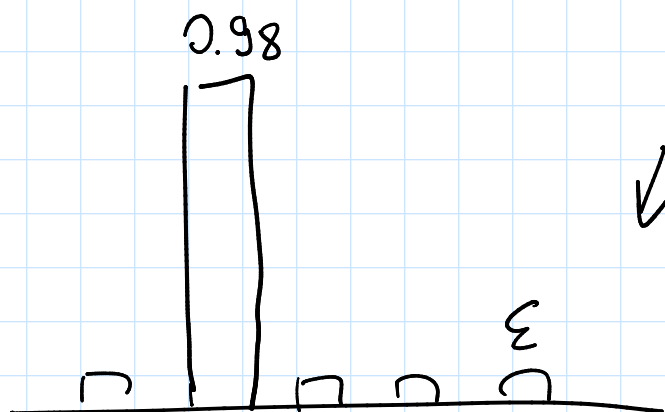
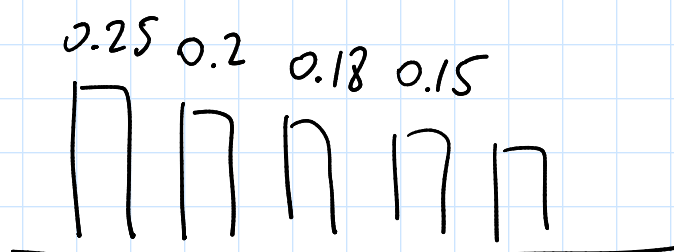
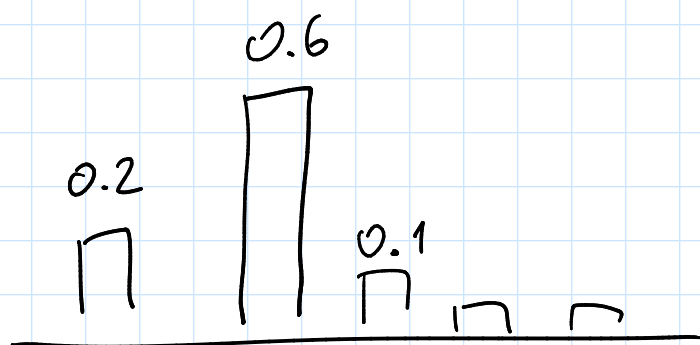
- Calibration & uncertainty estimation
- Distillation
- Pruning
- Quantization

Calibration

$$p_i = \frac{\exp(x_i)}{\sum_{k=1}^C \exp(x_k)}$$

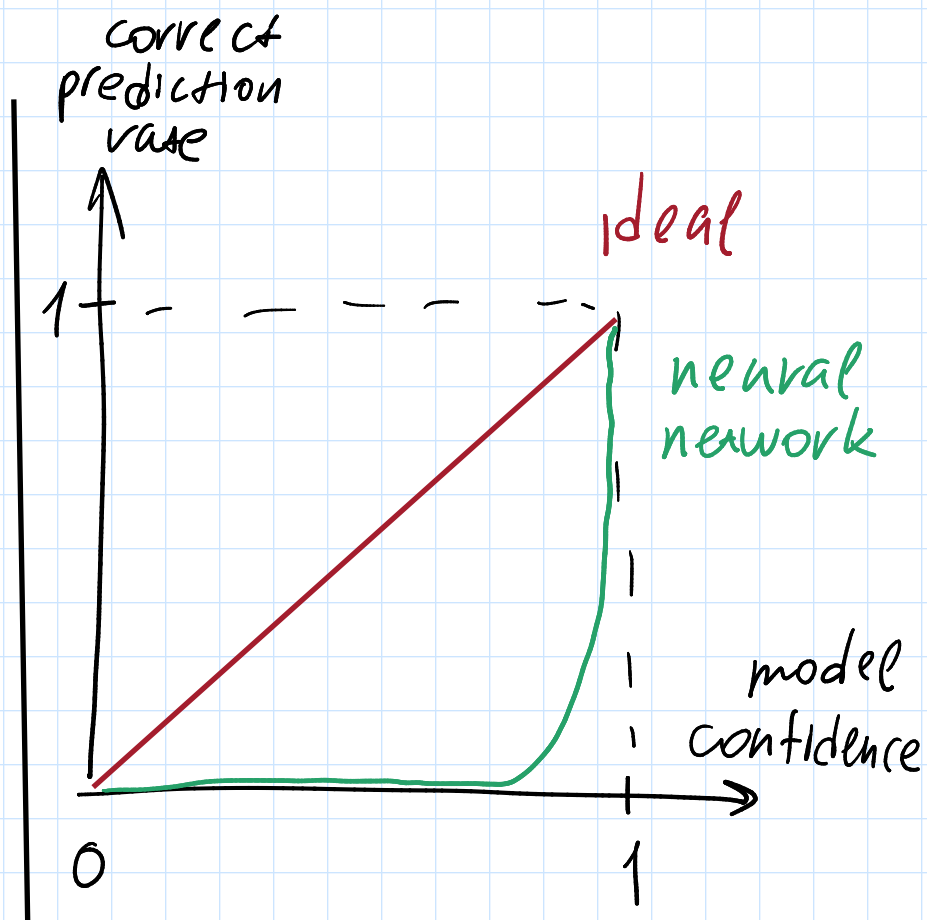
x - logits

p - probabilities



unconfident prediction

reality



- Temperature softmax ($\tau > 0$)

$$p_i = \frac{\exp(x_i / \tau)}{\sum_{k=1}^c \exp(x_k / \tau)}$$

$$1) \tau \rightarrow 0 \quad p \rightarrow \delta(\arg \max x) = (0, \dots, 0, \underset{i = \arg \max}{1}, \dots, 0)$$

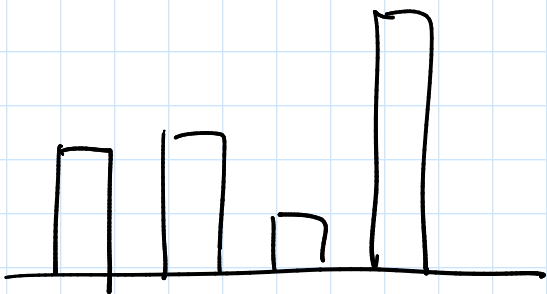
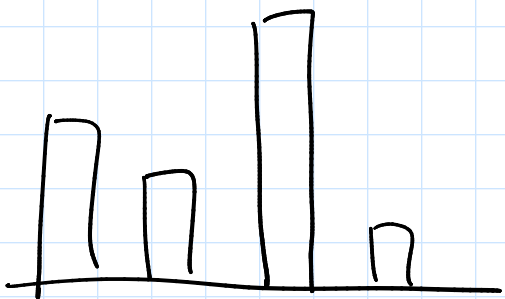
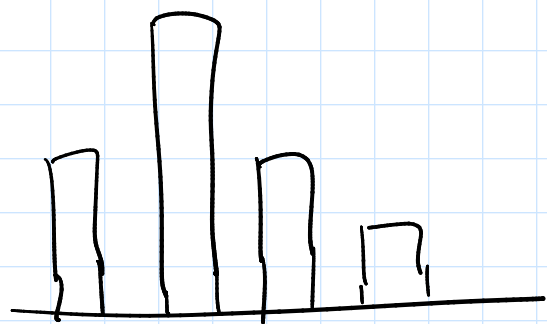
$$2) \tau \rightarrow \infty \quad p \rightarrow U(\{1, \dots, c\})$$

$\tau > 1$ usually in practice

$$\tau^* = \underset{\tau}{\operatorname{argmin}} \left(- \sum_{n=1}^N \sum_{k=1}^c [y_n = k] \log p_{nk}^{\tau} \right)$$

on validation set

- Ensemble (Deep Ensemble)



$$p_i^{\text{ens}} = \frac{1}{R} \sum_{r=1}^R p_i^r \quad (R=3)$$



(Knowledge) Distillation

P_{nk}^T — teacher probas
 P_{nk}^S — student probas

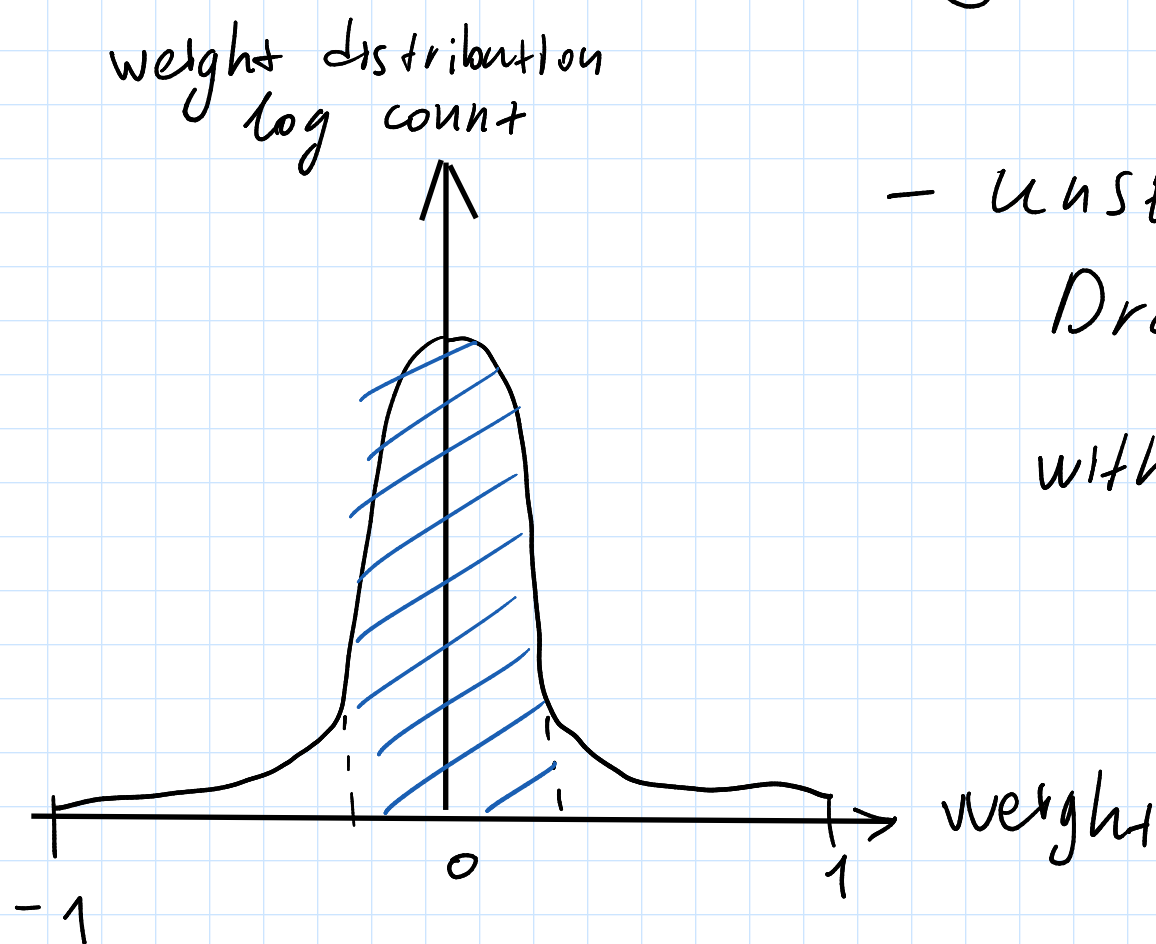
} with temperature τ

$$CE \text{ loss: } \mathcal{L}_{CE} = - \sum_{n=1}^N \sum_{k=1}^C [y_n = k] \log p_{nk}^S$$

$$\text{Distillation loss: } \mathcal{L}_D = - \sum_{n=1}^N \sum_{k=1}^C p_{nk}^T \log p_{nk}^S$$

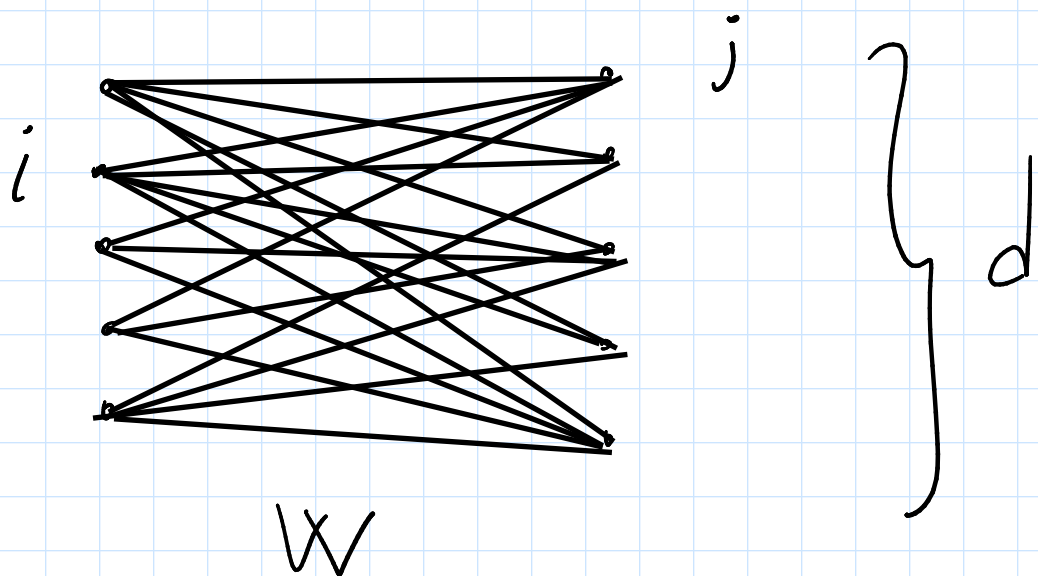
$$\mathcal{L} = \alpha \mathcal{L}_D + (1 - \alpha) \mathcal{L}_{CE}, \quad \alpha \in (0, 1)$$

Pruning



— unstructured pruning
Drop $\alpha\%$ weights
with lowest magnitude

- Structured pooling



$$p_i^\alpha = \sqrt[\alpha]{\sum_{j=1}^d |w_{ij}|^\alpha}$$

Staged pruning

1. Drop $\alpha\%$ weights
2. Fine-tune pruned network
3. Repeat

$$W \in \mathbb{R}^{d_1 \times d_2}$$

$$W = U \Sigma V^T - \text{SVD}$$

$$U \in \mathbb{R}^{d_1 \times s} \quad \Sigma \in \mathbb{R}^{s \times s}$$

$$V \in \mathbb{R}^{s \times d_2}$$

Quantization

training - float 32

Inference - int 8

$$W \in \mathbb{R}^{d_1 \times d_2}$$

- weight matrix

w - one scalar weight (float 32)

$$w = s_w (w^q - z_w)$$

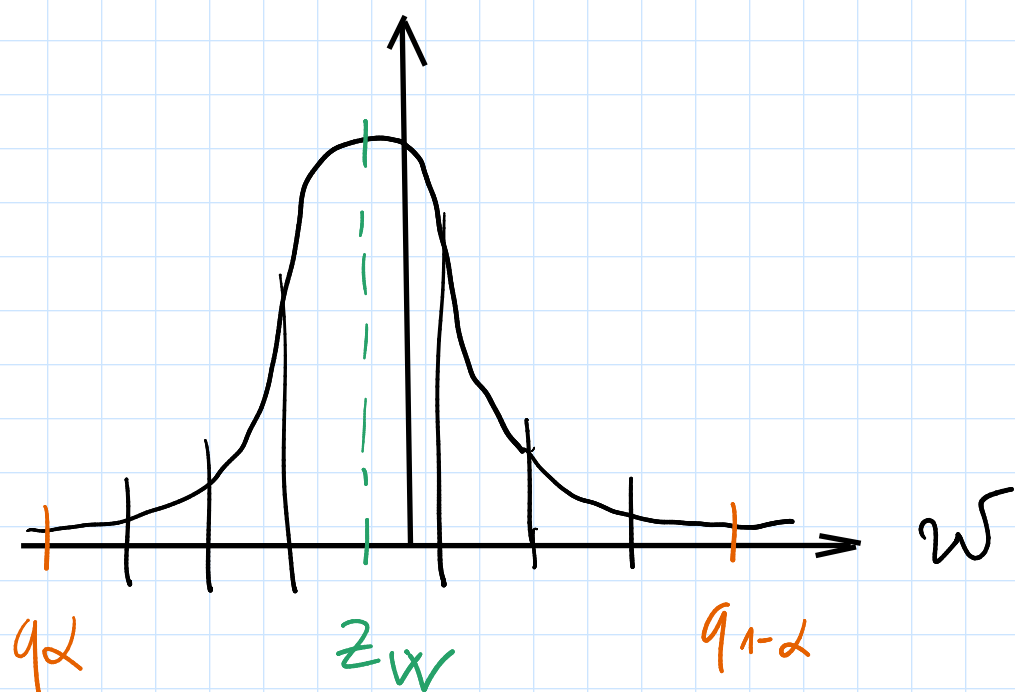
s_w - float32 - scale param

z_w - int8 - zero param

w^q - int8 - quantized weight

$$w^q = z_w + \frac{w}{s_w} \quad \text{float32}$$

$$w^q = \text{clip} \left(\text{round} \left(\frac{w}{s_w} \right) + z_w, -128, 127 \right)$$



$$z_w = \text{round} \left(\frac{q_{1-d} + q_2}{2} \right)$$

$$q_2 \rightarrow -128$$

$$q_{1-d} \rightarrow 127$$

$$X \in \mathbb{R}^{B \times d_1}$$

$$Y \in \mathbb{R}^{B \times d_2}$$

$$W \in \mathbb{R}^{d_1 \times d_2}$$

$$Y = XW$$

$$y_{ij} = \sum_{k=1}^{d_1} x_{ik} w_{kj}$$

$$s_x, z_x$$

$$s_y, z_y$$

$$s_w, z_w$$

$$\begin{aligned}
 y_{ij} &= \sum_{k=1}^{d_1} S_x (x_{ik}^q - z_x) \cdot S_w (w_{kj}^q - z_w) = \\
 &= S_x S_w \left[\underbrace{\sum_{k=1}^{d_1} x_{ik}^q w_{kj}^q}_{+ d_1 z_x z_w} - z_x \sum_{k=1}^{d_1} w_{kj}^q - z_w \sum_{k=1}^{d_1} x_{ik}^q \right] = S_y (y_{ij}^q - z_y)
 \end{aligned}$$

GPTQ — extra reading
 OBS → ORQ